

Patent source

2,572 collected
patent designs

Standardize and shortlist

1,253 patent seeds

Controlled perturbation

Sample a parent design

Perturb curvature,
thickness, materials,
or combinations

Record parent and
perturbation lineage

Physics reconstruction

Match target EFL

Evaluate semi-diameters
and place image plane

Reject material or
geometric failures

CodeV evaluation

Evaluate accepted
prescriptions in CodeV

Retain successful
ray traces

Record optical metrics

Dataset assembly

1,125,080 accepted
synthetic rows

Combine with 1,253
patent rows

Patent-only, 250k,
500k, and full variants